The Inheritance of Sex

Exam code: 0610 & 0970

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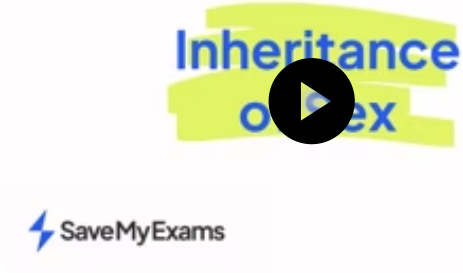

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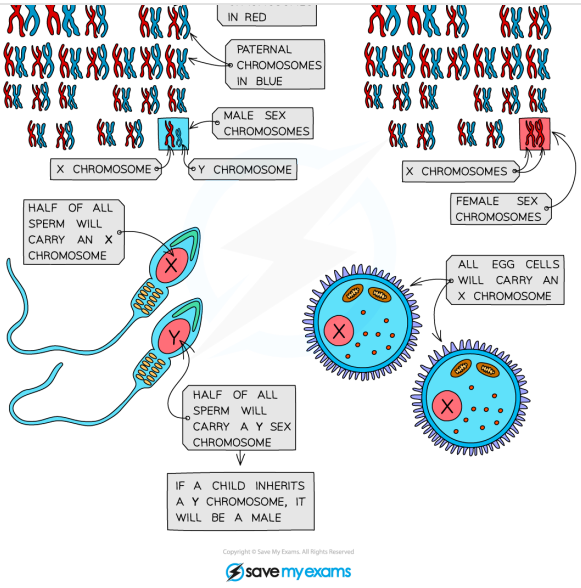
XX & XY Chromosomes

- Sex is determined by an **entire chromosome pair** (as opposed to most other characteristics that are just determined by one or a number of genes)
- **Females** have the sex chromosomes **XX**
- **Males** have the sex chromosomes **XY**
- As only a father can pass on a Y chromosome, he is **responsible for determining the sex of the child**
- He does this because
  - He produces (ejaculates) around 250 million sperm cells during sexual intercourse
  - Of those, half (125 million sperm) will be carrying his X chromosome
    - If one of these sperm fertilises the egg, the fetus will be female
  - The other 125 million of his sperm will be carrying his Y chromosome
    - Which will result in a male fetus if one of these fertilises the egg

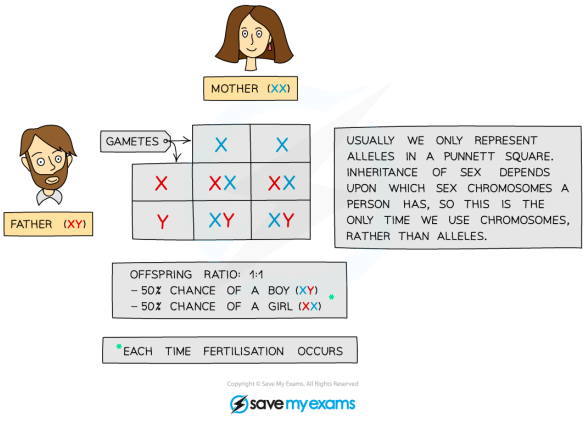
Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it



Sperm cells determine the sex of offspring

- The inheritance of sex can be shown using a **genetic diagram** (known as a **Punnett square**), with the X and Y chromosomes taking the place of the alleles usually written in the boxes



Punnett square showing the inheritance of sex

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